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Subject: IDS Roll No.: 63 Exp. No.: 2
Name of the Experiment:

Marks	Teacher's Signature with date
10	

Performed on:
Submitted on:

IDS Assignment - 2

1. Sequential organization & characteristics of linear data structures

it is a method of storing data elements in a continuos memory location where elememets in a continuos memory location elements are arranged one after another in a fixed order.

Characteristics :

- elements are stored in contigous memory locations.
- each element has a unique index or position.
- Direct access to elements is possible using indexing.
- Data is arranged in a linear sequence.
- size is usually fixed.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

2. Array as an abstract Data type (ADT).

ADT is collection of elements of the same data type stored in contiguos memory locations. & accessed using an index.

- Components of ADT :

→ Elements : Data items of the same type.
→ Index : position used to access elements.
→ Size : Total number of elements.

- Basic operations on array :

→ Transversal : visiting each element.
→ Insertion : Adding a new element.
→ Deletion : Removing an element.
→ Searching : finding an element.

3. Insertion, deletion & searching in a 1D array.

consider an array :

A = [10, 20, 30, 40]

insertion : insert is an position 2 :

- Shift elements right → A = [10, 20, 25, 30, 40]

- Deletion : Delete element at position 3 :

  shift elements left → A = [10, 20, 25, 40]

- searching : To search 25

  compare sequentially until found.

  25 is found in index 2.

4. Advantages & limitations of arrays

Advantages :

- fast direct access using index (O(1))
- simple & easy to implement.
- effeciently memory usage for fixed size data.
- good for storing homogeneous data.

Limitations :

- Fixed size.
- insertion & deletion are time-constant.
- wastage of memory if size not usefull.
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Effeciently comparision :

- insertion : O (n) due to shifting.
- Deletion : O(n) due to shifting.
- searching : O(n) for linear search
  O(logn) for binary search (sorted array).

5. 2- Dimentional array for student academic data :

2D array : Rows represent students.
Coulumns represent marks / subject.

Ex :-

Student	Math	science	English
S_1	85	90	88
S_2	78	82	80

Accessing data :

- marks [0][1] → science marks
- Tranversal uses nested loops.

Effecient manipulation :

- Direct indexing allows fast access
- Nested loop helps in proccesing Total & Averages.

Extension to n-dimentional arrays :

- Image proccesing (3D arrays for RGB)
- scientific simulations
- ML & data analysis.
- game development.